

1.5.7.6.1.1. AcceptOnlyWhenOn Bit

This bit specifies whether the OnWithTimedOff command is to be processed unconditionally or only when the OnOff attribute is equal to TRUE. If this sub-field is set to 1, the OnWithTimedOff command SHALL only be accepted if the OnOff attribute is equal to TRUE. If this sub-field is set to 0, the OnWithTimedOff command SHALL be processed unconditionally.

1.5.7.6.2. OnTime Field

This field is used to adjust the value of the OnTime attribute.

1.5.7.6.3. OffWaitTime Field

This field is used to adjust the value of the OffWaitTime attribute.

1.5.7.6.4. Effect on Receipt

On receipt of this command, if the AcceptOnlyWhenOn sub-field of the OnOffControl field is set to 1, and the value of the OnOff attribute is equal to FALSE, the command SHALL be discarded.

If the value of the OffWaitTime attribute is greater than zero and the value of the OnOff attribute is equal to FALSE, then the server SHALL set the OffWaitTime attribute to the minimum of the OffWaitTime attribute and the value specified in the OffWaitTime field.

In all other cases, the server SHALL set the OnTime attribute to the maximum of the OnTime attribute and the value specified in the OnTime field, set the OffWaitTime attribute to the value specified in the OffWaitTime field and set the OnOff attribute to TRUE.

If the values of the OnTime and OffWaitTime attributes are both not equal to 0xFFFF, the server SHALL then update these attributes every $1/10^{\text{th}}$ second until both the OnTime and OffWaitTime attributes are equal to 0, as follows:

- If the value of the OnOff attribute is equal to TRUE and the value of the OnTime attribute is greater than zero, the server SHALL decrement the value of the OnTime attribute. If the value of the OnTime attribute reaches 0, the server SHALL set the OffWaitTime and OnOff attributes to 0 and FALSE, respectively.
- If the value of the OnOff attribute is equal to FALSE and the value of the OffWaitTime attribute is greater than zero, the server SHALL decrement the value of the OffWaitTime attribute. If the value of the OffWaitTime attribute reaches 0, the server SHALL terminate the update.

1.5.8. State Description

The operation of the On/Off cluster with respect to the On, Off, and OnWithTimedOff commands is illustrated in [On/Off Cluster Operation State Machine](#). In this diagram, the values X and Y correspond to the OnTime and OffWaitTime fields, respectively, of the OnWithTimedOff command. In the Timed On state, the OnTime attribute is decremented every $1/10^{\text{th}}$ second, unless its value equals 0xFFFF. Similarly, in the Delayed Off state, the OffWaitTime attribute is decremented every $1/10^{\text{th}}$ second, unless its value equals 0xFFFF.